BIE-MA1 : sample assessment test - instructions

- You have 60 minutes to complete the test. Calculators and other electronic devices are not allowed.
- It is not allowed to use cell-phone or to communicate with anyone except the examiner.
- Not only the result, but even **correct procedure and its understandability** is required. Always write, what you are attempting to do, so it is clear that you know what to do, even if you do it wrong.
- Always write and explain procedures you used to get to the result, i.e., theorems you used, ideas behind, etc.

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Summer 2021/2022 BIE-MA1 : assesment test-sample					Test 1S
Family name, Given name	1	2	3	4	**Sum**

Problem 1. (9 points) Decide whether it is true or false that

a) $\frac{2}{n^2} + \frac{4}{\sqrt[3]{n}} = \mathcal{O}\left(\frac{1}{\sqrt[4]{n}}\right)$

b) $\frac{3}{\sqrt[3]{n}} = o\left(\frac{1}{n^2}\right)$

c) $\frac{2}{n^2} + \frac{4}{\sqrt[3]{n}} = o\left(\frac{1}{\sqrt[4]{n}}\right)$

Prove your statements!

Problem 2. (8 points) Consider the sequence defined by

$$a_n = \frac{1}{3n-5}$$

a) Is the sequence (eventually) monotonic? If yes, specify the type of monotonicity (increasing, strictly increasing, decreasing,...).

b) Is the sequence bounded from below, bounded from above, or bounded (from both sides)? Find the appropriate bounds.

Problem 3. (12 points) Compute the following limits of sequences (or if the limit does not exist, explain why):

a)

$$\lim_{n\to+\infty} \frac{\ln(n^3+2n+1)}{\ln(2n^5+n)}.$$

b)

$$\lim_{n\to+\infty} \frac{(-1)^n n^2 + \sqrt[4]{n} + \cos(n^{10}) + 3^{-n}}{n! + n^2}.$$

Problem 4. (10 points) For which values of the parameter $a, b \in \mathbb{R}$ is the function

$$f(x) = \begin{cases} \frac{b\tan(4-x^2)}{(x-2)}, & x > 2 \\ a, & x = 2 \\ \arcsin(x-2) & x < 2 \end{cases}$$

continuous at $x = 2$?

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